

Name: _____

Class: _____

Date: _____

Lab 5: Debugging Detective

Python Lab · Section 1.4 Identifying & Correcting Errors



Goal

Find and fix syntax, run-time, and logic errors using test cases, tracing, and extra print statements.

Warm-up: Predict

```
x = 10
y = 0
print(x / y)
```

Will this run? What kind of error happens and when?

AP Connection

- Syntax, run-time, logic, and overflow errors (1.4).
- Test with boundary values!

Build It

Step 1. CASE 1 — fix it and name the error type:

```
print("Welcome!")
name = input("Name: ")
print("Hi " + name)
```

Step 2. CASE 2 — should print the average of 3 test scores. Which input reveals the bug?

```
a = 90
b = 80
c = 70
avg = a + b + c / 3
print(avg)
```

Step 3. CASE 3 — should print "Teen" for ages 13 to 19. Write 4 test cases (including boundaries) and fix it.

```
age = int(input("Age: "))
if age > 13 and age < 19:
    print("Teen")
```

Step 4. CASE 4 — should add up 1 through 5 (answer 15). Add print statements inside the loop to find the bug.

```
total = 0
for i in range(1, 5):
    total = total + i
print(total)
```

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Build It (continued) · Level Up

Build It (continued)

Step 5. CASE 5 — crashes for some inputs. Find one that crashes it.

```
items = ["pen", "book", "bag"]
n = int(input("Item number (1-3): "))
print(items[n])
```

Level Up Challenges

Mild Make a table: case number, error type, fix.

Medium Write a buggy program for a partner. They name and fix the error.

Spicy Case 5 has TWO problems (Python lists start at 0, and no check for invalid input). Fix both.

Python → AP Pseudocode

```
items = ["pen", "book", "bag"]
print(items[0])
```

Python's `items[0]` is which index in AP pseudocode? Write the DISPLAY statement.

Exit Ticket

Why are boundary values (like 13 and 19) the best test cases?
